Part IA Math methods for NST and CompSci

Lent term 2026, course B (24 lectures)

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they draw on previous notes of Natalia Berloff.

Introduction

The purpose of these notes

These notes give a self-contained presentation of the material in this course. There will also be handwritten notes in lectures: they will focus on key ideas and may skip some of the details that are given here.

It is strongly recommended that you take notes in lectures, and use those handwritten notes in conjunction with these ones, in order to learn the material and attempt the examples sheets.

If you find a mistake in these notes then: Please check moodle to see if this has already been noted. If not, please email `rlj22@cam.ac.uk` so that it can be fixed.

The purpose of this course

We aim to teach you the methods for solving practical problems. In doing that, we will explain how (and why) the methods work, as well as how to operate them

These lectures will ensure that you are equipped to take the exams, but (we believe that) a more important goal is that you will have the right skills to make progress in later parts of your degree. This means that "why do we have to learn this?" might be an interesting question to ask the lecturer, but "will this be in the exam?" is (usually) not an interesting question.

One other note on exams:

The "syllabus" for exam questions is given by the official schedules. If you are familiar with these notes and with the ideas that appear in the problem sheets, you will have seen all the material that is given in the schedules. The handwritten notes from lectures will illustrate the key points from these notes.

In the notes, some extra information is given in footnotes[1], including additional definitions, and explanations of how some results might be derived. These are for your interest and are not essential for the course.

[1] This is a footnote.

1 Ordinary differential equations

1.1 Introduction

A ordinary differential equation (ODE, sometimes called just a "differential equation") is an equation that involves an unknown function and its derivatives. They appear in many different scientific contexts where the unknown function would usually be some physical quantity. This quantity is called the **dependent variable**, it is a function of the **independent variable**.

Example 1 (from physics)**.** *A particle has velocity v which is a function of time t. It feels a gravitational force downwards ($-mg$, where m is mass and g the field strength) and a drag force $-\mu v^3$ (in the opposite direction to the velocity, where μ is a constant). Then Newton's 2nd law ("mass $\times$ acceleration $=$ total force") says that*

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = -mg - \mu v^3 \tag{1.1}$$

If we can solve this equation then we obtain the velocity v as a function of time.

In the above example, v is the dependent variable and t is the independent variable. The first derivative of v with respect to t is written as

$$\frac{\mathrm{d}v}{\mathrm{d}t} \qquad \text{or} \qquad v'(t)$$

For most of our discussion we will use y for the dependent variable and x for the independent variable. In general we write things like

$$\frac{\mathrm{d}y}{\mathrm{d}x}, \qquad \frac{\mathrm{d}^2y}{\mathrm{d}x^2}, \quad \ldots \quad \frac{\mathrm{d}^ny}{\mathrm{d}x^n} \qquad \text{or} \quad y'(x), \qquad y''(x), \quad \ldots \qquad y^{(n)}(x) \ .$$

Definition 1 (Order of an ODE)**.** *A* **first-order** *differential equation includes the first derivative of the dependent variable but no higher derivatives. A* **second-order** *differential equation includes the second derivative and (maybe) the first derivative, but no higher derivatives. An n***th-order** *equation incudes the nth derivative but no higher derivatives.*

For example, Eq. (1.1) is a first-order ODE. An example of a second-order ODE would be $y''(x) + 4y'(x) + 3y(x) = 0$.

Solving an ordinary differential equation (ODE) amounts to working out the dependent variable in terms of the independent variable. In the simplest case this would be an explicit formula for the dependent variable. In Example 1, this would be a expression for the velocity as a function of time. However, such **analytical** solutions are not possible in general. If we can't make progress analytically then we can usually obtain **numerical** solutions, which means that a computer can calculate the dependent variable. We focus in these lectures on cases where we can obtain analytical solutions, which illustrate the general mathematical principles.

1.2 First order ODEs

We are going to discuss various kinds of ODEs that we are able to solve.

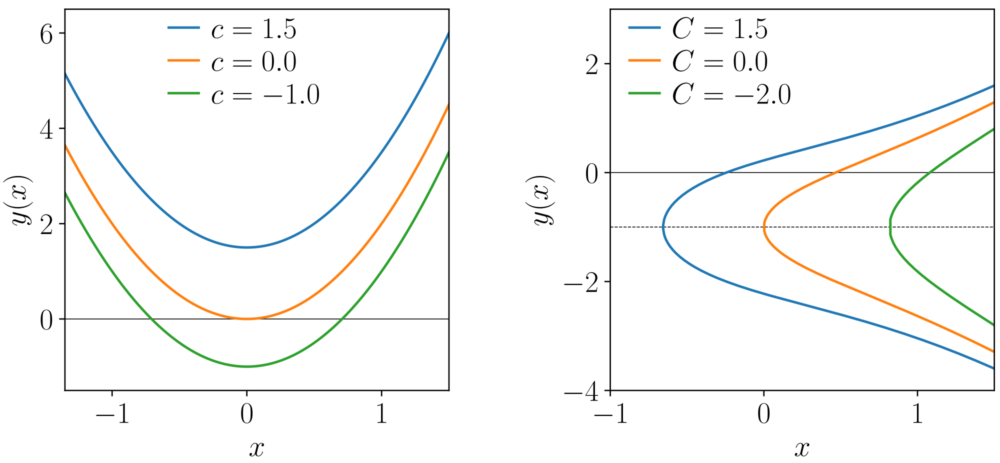

Figure 1.1: Examples of particular solutions. (Left) For the integrable ODE $\frac{dy}{dx} = 4x$ the general solution is $y(x) = 2x^2 + c$. By choosing values for c we can plot some particular solutions, these differ only by constant shifts. (Right) For the separable ODE of Example 2 the general solution is Eq. (1.8). The particular solutions have different shapes, note we can take either sign of the square root so y can take two possible values.

1.2.1 Integrable equations (general solution)

Definition 2. *An* **integrable ODE** *can be written in the form*

$$\frac{dy}{dx} = f(x) \tag{1.2}$$

for some function $f = f(x)$.[1] *Note the right hand side does not depend on y.*

Integrating both sides of Eq. (1.2) gives

$$y = \int f(x)dx$$

Suppose we can find a function $F = F(x)$ such that $F'(x) = f(x)$. This amounts to "doing the integral"[2] and so

$$y(x) = F(x) + c \tag{1.3}$$

is the solution to Eq. (1.2), where c is an integration constant. We can choose any value of c and we still have a valid solution to the ODE. In other words, that there are "infinitely many solutions" to the ODE.

Eq. (1.3) is called the **general solution** of Eq. (1.2) because it contains all the possible solutions. If we take particular values for c then we have **particular solutions**. See Fig. 1.1 (left panel).

1.2.2 Separable equations

Definition 3. *A* **separable ODE** *takes the form*

$$\frac{dy}{dx} = \frac{f(x)}{g(y)} \tag{1.4}$$

[1]We use the notation "$f = f(x)$" to indicate that f is a function of x. For rigorous mathematicians then there would be other ways to write this, but the current version is unambiguous and clear enough for our purposes.

[2]It is tempting to say that "F is the integral of f" but that is not correct because the (indefinite) integral of f has to include the integration constant, it would be $F(x) + c$. Some people would call F the "anti-derivative" of f.

where f, g are two given functions.

This cannot be solved in the simple way that we used for integrable ODEs. Instead we multiply both sides by g and integrate to get

$$\int g(y)\frac{dy}{dx}dx = \int f(x)dx$$

Remember that y is itself a function (of x). Hence we can do a substitution (that is, a change of integration variable) on the left hand side; we get

$$\int g(y)dy = \int f(x)dx \tag{1.5}$$

To get from Eq. (1.4) to Eq. (1.5), it is common to write the intermediate step $g(y)\mathrm{d}y = f(x)\mathrm{d}x$ where it looks like we "multiplied by both sides of Eq. (1.4) by $g(y)$ and then also multiplied by $\mathrm{d}x$". This is called "separating the variables" because we end up with a LHS that only depends on y and a RHS that depends only on x. Of course $\frac{\mathrm{d}y}{\mathrm{d}x}$ is not a fraction so we can't really multiply by $\mathrm{d}x$, but the method does work for solving the ODE. We will have more to say about this in later parts of the course (for example Sec. 2.2.3).

Now suppose (similar to Sec. 1.2.1) that we can find functions $F = F(x)$ and $G = G(y)$ such that $F'(x) = f(x)$ and $G'(y) = g(y)$. Then we can do the integrals in Eq. (1.5) to get

$$G(y) = F(x) + c \tag{1.6}$$

This is a general solution to the ODE (1.4). Only one constant of integration is needed here (we could have written $G(y) + c_1 = F(x) + c_2$ but since c_1, c_2 are both arbitrary constants then we combine them into a single constant $c = c_2 - c_1$).

Example 2 (separable ODE)**.** *Solve*

$$\frac{dy}{dx} = \frac{x^2+1}{y+1} \tag{1.7}$$

Multiply both sides by $(y+1)$ and integrate:

$$\int (y+1)\frac{dy}{dx}dx = \int (x^2+1)dx$$

Changing integration variable gives $\int (y+1)dy = \int (x^2+1)dx$ and integrating both sides gives

$$\frac{y^2}{2} + y = \frac{x^3}{3} + x + c$$

which is the general solution. This is a quadratic in y so we can go one step further and write y directly as a function of x:

$$y(x) = -1 \pm \sqrt{(2x^3/3) + 2x + C} \tag{1.8}$$

where now $C = 2c + 1$ is the integration constant. (Since c was an arbitrary constant, we are free to use C instead.) See Fig. 1.1 (right panel).

We should always check our solutions by substituting back into the ODE. In this case

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= \frac{x^2+1}{\sqrt{(2x^3/3) + 2x + C}} \\ &= \frac{x^2+1}{y+1}\end{aligned}$$

where the first "=" uses chain rule on Eq. (1.8) and the second uses the same equation to write the denominator in terms of y. Hence our solution does indeed obey Eq. (1.7).

The previous example is nice because we can obtain y explicitly as a function of x. The next example shows that this is not always the case.

Example 3 (another separable ODE)**.**

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x^2+1}{1+\cos y}$$

Separate variables to get $\int(1+\cos y)\,\mathrm{d}y = \int(x^2+1)\,\mathrm{d}x$ *and integrate to get*

$$y + \sin y = (x^3/3) + x + c \tag{1.9}$$

This is the general solution.

Note the general solution does not give y *explicitly in terms of* x*, indeed this is not possible here. However, we have characterised all possible solutions for* y*. (If we wanted to plot the solutions then we could solve Eq. (1.9) numerically.)*

1.2.3 Geometric interpretation of first-order ODEs

Our examples of first-order ODEs can all be written as

$$\frac{\mathrm{d}y}{\mathrm{d}x} = F(x,y)$$

Remember that our general solution has a constant of integration, so there is an infinite family of particular solutions to this equation.

There is a *geometrical interpretation* of such ODEs: For every point (x,y) where $F(x,y)$ is well-defined, there is one particular solution passing through that point, and F is the gradient of the solution. This geometrical interpretation holds even if there is no explicit solution to the ODE. For the right graph in Fig. 1.1, the gradient of the line passing through (x,y) is given by Eq. (1.7) as $\frac{x^2+1}{y+1}$.

1.2.4 Linear equations and integrating factors

So far we dealt with integrable and separable ODEs [Eqs. (1.2) and (1.4)]. Another interesting type of first-order ODE takes the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} + p(x)y = f(x) \tag{1.10}$$

This is called a **linear** ODE. An important special case is where $f=0$, this is called a **homogeneous** linear ODE. If $f \neq 0$ then we have an **inhomogeneous** linear ODE. These terms are explained in more detail later (Sec. 1.3.1).

For the homogeneous case, the ODE can be put into separable form as in Eq. (1.4). That is

$$\frac{\mathrm{d}y}{\mathrm{d}x} + p(x)y = 0 \qquad \text{is the same as} \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{-p(x)}{1/y}\ . \tag{1.11}$$

From here we get

$$\int \frac{\mathrm{d}y}{y} = -\int p(x)\,\mathrm{d}x\ .$$

If we can find $P = P(x)$ such that $P'(x) = p(x)$ then we can integrate both sides to get $\ln y = -P(x) + c$. Then taking exponential of both sides we get

$$y(x) = A\mathrm{e}^{-P(x)} \tag{1.12}$$

where $A = \mathrm{e}^c$ is an arbitrary constant. This is the general solution of the homogeneous equation.

A different way to get this result is to multiply Eq. (1.11) through by $\mathrm{e}^{P(x)}$ which gives

$$\mathrm{e}^{P(x)}\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{e}^{P(x)}p(x)y = 0 \tag{1.13}$$

and note that $\mathrm{e}^{P(x)}p(x) = \frac{\mathrm{d}}{\mathrm{d}x}\mathrm{e}^{P(x)}$ by chain rule [remember $p(x) = P'(x)$]. Then use product rule to write Eq. (1.13) as

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(y\mathrm{e}^{P(x)}\right) = 0 \tag{1.14}$$

Integrating both sides with respect to x gives $y\mathrm{e}^{P(x)} = A$ where A is constant of integration, this is the same as Eq. (1.12). The object $\mathrm{e}^{P(x)}$ is called an **integrating factor**.

This last method shows how to solve the inhomogeneous equation (1.10). Multiply that equation by $\mathrm{e}^{P(x)}$, the left hand side is the same as the homogeneous equation so

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(y\mathrm{e}^{P(x)}\right) = f(x)\mathrm{e}^{P(x)}\ . \tag{1.15}$$

Integrating both sides we get

$$y(x)\mathrm{e}^{P(x)} = \int f(x)\mathrm{e}^{P(x)}\,\mathrm{d}x\ . \tag{1.16}$$

It only remains to do the integral on the right hand side and then we have our (general) solution. We illustrate this with an example.

Example 4 (linear first-order ODE)**.** *Solve*

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 3y = \mathrm{e}^{-2x} \tag{1.17}$$

We see that $p(x) = 3$ and $q(x) = \mathrm{e}^{-2x}$. Taking $P(x) = 3x$ we have $P'(x) = p(x)$, as required so the integrating factor is e^{3x}. Multiply the equation by the integrating factor

$$\mathrm{e}^{3x}\frac{\mathrm{d}y}{\mathrm{d}x} + 3\mathrm{e}^{3x}y = \mathrm{e}^{x}$$

and use product rule to write this as $\frac{\mathrm{d}}{\mathrm{d}x}(ye^{3x}) = e^x$. Integrate both sides to get $ye^{3x} = e^x + A$. Rearranging we finally obtain

$$y(x) = \mathrm{e}^{-2x} + A\mathrm{e}^{-3x} \tag{1.18}$$

We should check that this solves Eq. (1.17): we have

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} + 3y &= (-2\mathrm{e}^{-2x} - 3A\mathrm{e}^{-3x}) + 3(\mathrm{e}^{-2x} + A\mathrm{e}^{-3x})\\ &= \mathrm{e}^{-2x}\end{aligned}$$

as required.

1.2.5 General solution and boundary conditions

We have seen that general solutions to our ODEs contain constants of integration. In practical applications, we usually need to fix these constants. This requires additional information, beyond the ODE itself. This information comes in the form of one or more boundary conditions.

For the (first-order) ODEs considered so far then we have one constant of integration. To fix this constant, we need to know the actual value of the dependent variable, for some specific value of the independent variable.

Important fact: For an nth order ODE we expect our general solution to contain n integration constants. To fix these we will need n pieces of information about the dependent variable.

In Example 1, the natural boundary condition would be to specify the velocity $v(0)$ at time $t = 0$. (This might also be called an *initial condition*.) Then the velocity can be determined for all subsequent times, by Newton's laws.

Example 5 (boundary conditions). *Recall Example 2 was* $\frac{dy}{dx} = \frac{x^2+1}{y+1}$. *The general solution was*

$$y(x) = -1 \pm \sqrt{(2x^3/3) + 2x + C} \tag{1.19}$$

We take as boundary condition $y(0) = 1$. *Plugging* $(x, y) = (0, 1)$ *into the general solution we get* $1 = -1 \pm \sqrt{C}$. *We must take the* $+$ *sign and* $C = 4$. *Our particular solution is then*

$$y(x) = -1 + \sqrt{(2x^3/3) + 2x + 4}$$

The function y *is now fixed (there are no arbitrary constants or signs left over).*

1.2.6 Use of substitutions

We have considered three types of ODE where we can *always* obtain the solution by doing integrals (integrable, separable, linear). There are some other cases where we can make progress by a method based on substitution (or change of variables). We illustrate this with some specific examples.

An interesting type of equation is

$$\frac{\mathrm{d}y}{\mathrm{d}x} = f\left(\frac{y}{x}\right) \tag{1.20}$$

where f is some function. [This is called a "homogeneous ODE", but note the meaning of "homogeneous" in this context is a bit different to the homogeneous linear equations discussed previously(!).] It is natural to introduce a new function $u(x) = y(x)/x$ so that the right hand side becomes $f(u)$. Noting that $y(x) = xu(x)$ we use product rule to get

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}(xu) \\ &= u + x\frac{\mathrm{d}u}{\mathrm{d}x}\end{aligned}$$

Then Eq. (1.20) can be written as

$$u + x\frac{\mathrm{d}u}{\mathrm{d}x} = f(u) \tag{1.21}$$

and so

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{f(u) - u}{x} \tag{1.22}$$

and this is of separable type [Eq. (1.4), with dependent variable u]. Hence we can solve it using separation of variables.

Example 6 (substitution $u = y/x$). *Consider*

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x^2 + y^2}{xy} \qquad \textit{with initial condition} \quad y(1) = 1$$

Note that the right hand side is $(x/y)+(y/x)$ so this has the form of Eq. (1.20). Let $y(x)=xu(x)$, we get

$$u+x\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{1}{u}+u$$

so

$$\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{1}{ux}$$

This is of separable type, from which we obtain $\int u\,\mathrm{d}u=\int\frac{\mathrm{d}x}{x}$. Hence the general solution is $\frac{1}{2}u^2=\ln x+c$. which we write as $u(x)=\pm\sqrt{2\ln x+2c}$.

The initial condition $y(1)=1$ means that $u(1)=1$ so we have to take the positive square root and $c=\frac{1}{2}$. So finally $u=\sqrt{1+2\ln x}$, and putting things back in terms of y we have

$$y(x)=x\sqrt{1+2\ln x}$$

We check this works by writing

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x}&=\sqrt{1+2\ln x}+x\frac{(1/x)}{\sqrt{1+2\ln x}}\\&=\frac{y}{x}+\frac{x}{y}\end{aligned}$$

which is indeed the same as $(x^2+y^2)/(xy)$, as required.

Another interesting example is the **Bernoulli equation**:

$$\frac{\mathrm{d}y}{\mathrm{d}x}+p(x)y=q(x)y^n$$

If $n=0$ or $n=1$ then you can check that this equation reduces to linear type and we can already solve it. In other cases we should change variables as $z(x)=y(x)^{1-n}$. Then $\frac{\mathrm{d}z}{\mathrm{d}x}=(1-n)y(x)^{-n}\frac{\mathrm{d}y}{\mathrm{d}x}$. Using this substitution, the Bernoulli equation becomes

$$\frac{\mathrm{d}z}{\mathrm{d}x}=(1-n)y^{-n}[q(x)y^n-p(x)y]$$

which can be rearranged into linear form as

$$\frac{\mathrm{d}z}{\mathrm{d}x}+(1-n)p(x)z=(1-n)q(x)$$

This we can solve by integrating factor.

Example 7 (A Bernoulli equation)**.** *Solve*

$$\frac{\mathrm{d}y}{\mathrm{d}x}+xy=x^3y^2$$

This of the Bernoulli form with $n=2$ so let $z(x)=\frac{1}{y(x)}$ and $\frac{\mathrm{d}z}{\mathrm{d}x}=\frac{-1}{y^2}\frac{\mathrm{d}y}{\mathrm{d}x}$. Dividing our ODE by $-y^2$ and using these results, we get

$$\frac{\mathrm{d}z}{\mathrm{d}x}-xz=-x^3$$

which is a linear ODE for z. Comparing with Eq. (1.10) we have $p(x)=-x$ so we use integrating factor $\mathrm{e}^{-x^2/2}$. We get

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(z\mathrm{e}^{-x^2/2}\right)=-x^3\mathrm{e}^{-x^2/2}$$

We want to integrate both sides, for the right hand side we need integration by parts. We get

$$\begin{aligned} z\mathrm{e}^{-x^2/2} &= \int x^2 \left(-x\mathrm{e}^{-x^2/2}\right) \mathrm{d}x \\ &= x^2\mathrm{e}^{-x^2/2} - \int 2x\mathrm{e}^{-x^2/2}\mathrm{d}x \\ &= x^2\mathrm{e}^{-x^2/2} + 2\mathrm{e}^{-x^2/2} + c \end{aligned}$$

So finally $z = x^2 + 2 + c\,\mathrm{e}^{x^2/2}$ *which means*

$$y(x) = \frac{1}{x^2 + 2 + c\,\mathrm{e}^{x^2/2}} \tag{1.23}$$

We should check that this works by substituting back in, this is left as an exercise.

Some other examples of substitutions are explored on the problem sheet.

1.2.7 Application : radioactive decay

We illustrate the use of first-order ODEs in an example that is relevant in several different areas of the physical sciences: radioactive decay.

General equations

Suppose that radioactive isotope A decays to some new isotope B with rate k_a and then B decays to isotope C with rate k_b. Let $a(t), b(t), c(t)$ be the numbers of atoms of A,B,C respectively.

The independent variable will be t and the dependent variables will be a, b, c. We are given initial conditions $a(0) = a_0$, $b(0) = b_0$, $c(0) = c_0$ and the aim is to find a, b, c for all positive t. [In fact it will be enough to find only a, b.]

We know (from physics) that each atom decays independently. This means mathematically that

$$\frac{\mathrm{d}a}{\mathrm{d}t} = -k_a a, \qquad \frac{\mathrm{d}b}{\mathrm{d}t} = k_a a - k_b b, \qquad \frac{\mathrm{d}c}{\mathrm{d}t} = k_b b \tag{1.24}$$

The equation for $\frac{\mathrm{d}a}{\mathrm{d}t}$ does not contain b and it is of separable type. You may already know its solution but you can follow the above analysis to show that

$$a(t) = a_0\mathrm{e}^{-k_a t} \tag{1.25}$$

Putting this into the equation for $\frac{\mathrm{d}b}{\mathrm{d}t}$ and adding $k_b b$ to both sides, we obtain

$$\frac{\mathrm{d}b}{\mathrm{d}t} + k_b b = k_a a_0 \mathrm{e}^{-k_a t} \tag{1.26}$$

This is of linear type with $p(t) = k_b$ and $f(t) = k_a a_0 \mathrm{e}^{-k_a t}$. Hence we use integrating factor $\mathrm{e}^{k_b t}$ to obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}(b\mathrm{e}^{k_b t}) = k_a a_0 \mathrm{e}^{(k_b - k_a)t}$$

Integrate both sides:

$$b(t)\mathrm{e}^{k_b t} = \frac{k_a a_0 \mathrm{e}^{(k_b - k_a)t}}{k_b - k_a} + C$$

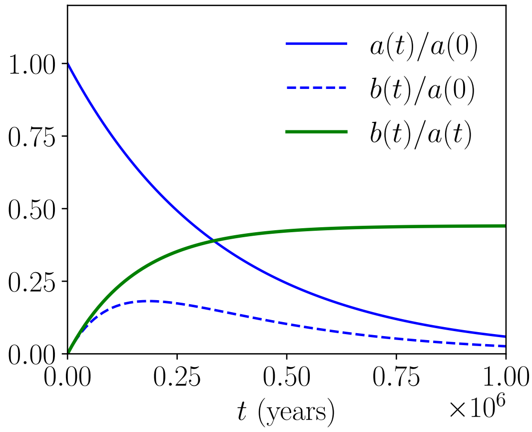

Figure 1.2: Results for radioactive decay. We show concentrations of Uranium-234 [which is $a(t)$] and Thorium-230 [which is $b(t)$] over a time period if 1 million years, these are normalised by the initial concentration of Uranium-234 [which is $a(0)$]. We also show the ratio $a(t)/b(t)$, which increases from zero towards a limiting value $\frac{k_a}{k_b-k_a}$.

where C is constant of integration. Multiplying both sides by $\mathrm{e}^{-k_b t}$ gives

$$b(t) = \frac{k_a a_0 \mathrm{e}^{-k_a t}}{k_b - k_a} + C\mathrm{e}^{-k_b t} \tag{1.27}$$

(Note: this formula does not apply in the special case $k_a = k_b$. That situation has to be analysed separately, it could be interesting to think about.)

We know $b(0) = b_0$ from which we find $C = b_0 - \frac{k_a a_0}{k_b - k_a}$ and so

$$b(t) = \frac{k_a a_0 \mathrm{e}^{-k_a t}}{k_b - k_a} + \left[b_0 - \frac{k_a a_0}{k_b - k_a}\right] \mathrm{e}^{-k_b t} \tag{1.28}$$

Finally the equation for $\frac{\mathrm{d}c}{\mathrm{d}t}$ is of integrable type and can be solved. (We don't write the answer here, note the answers for a, b don't depend on any properties of isotope C.)

Practical case

Consider the case where A is Uranium-234 and B is Thorium-230 and C is Radium-226. (No chemistry is required to understand this example, beyond that these are substances made of atoms!) The half-life of Uranium-234 is τ_a=245,000 years, while that of Thorium-230 Th is τ_b =75,000 years, these are related to rate constants as $k_a = \frac{\ln 2}{\tau_a}$ and $k_b = \frac{\ln 2}{\tau_b}$; hence $k_a < k_b$.

Organisms such as coral can take up uranium from seawater as they form, but they don't take up thorium (which is insoluble). Taking $t = 0$ as the time that the coral was formed, this means that $a_0 > 0$ but $b_0 = 0$. In this case we can use the solutions above to see that

$$\frac{b(t)}{a(t)} = \frac{k_a}{k_b - k_a}\left[1 - \mathrm{e}^{(k_a - k_b)t}\right]$$

This result is independent of a_0, so it only depends on k_a, k_b, t. (This ratio increases from zero at $t = 0$ to a positive limiting value at large times.)

The various quantities are plotted in Fig. 1.2.

Note that $b(t)/a(t)$ is the ratio of ^{230}Th to ^{234}U, this can be measured in coral samples today, we call it θ. Then we can rearrange the above equation to get

$$t = \frac{1}{k_a - k_b} \ln\left[1 - \left(\frac{k_b}{k_a} - 1\right)\theta\right]$$

This formula gives the elapsed time t since the coral was formed, in terms of known (or measured) quantities.

1.3 Second order ODEs (linear cases)

We considered several different types of first-order ODEs. At second order the situation is more complicated so we restrict to **linear** ODEs, we explain this terminology just below. For now we just state a general linear second-order ODE is

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + 2p(x)\frac{\mathrm{d}y}{\mathrm{d}x} + q(x)y = f(x) \tag{1.29}$$

where p, q, f are arbitrary functions. (The factor of 2 in front of p is arbitrary but will be useful later. If there was some function multiplying $\frac{\mathrm{d}^2 y}{\mathrm{d}x^2}$ then we should divide by this function to put the equation into the general form above.)

1.3.1 Homogeneous and inhomogeneous linear equations

It is sometimes useful to write Eq. (1.29) as

$$Ly = f \tag{1.30}$$

where

$$L = \frac{\mathrm{d}^2}{\mathrm{d}x^2} + 2p(x)\frac{\mathrm{d}}{\mathrm{d}x} + q(x) \tag{1.31}$$

is a linear differential operator. This operator acts on functions and produces new functions. (This is just like the operator "$\frac{\mathrm{d}}{\mathrm{d}x}$" which is already familiar to you.)

Definition 4. *A* **linear operator** *L has the following two properties:*

$$L(\alpha u) = \alpha L u \qquad L(u + v) = Lu + Lv \tag{1.32}$$

where α is an arbitrary real (or complex) number and u, v are arbitrary functions. A **linear differential operator** *is a linear operator that includes derivatives of u.*

You already know that the properties of Eq. (1.32) hold for the operators $\frac{\mathrm{d}^2}{\mathrm{d}x^2}$ and $\frac{\mathrm{d}}{\mathrm{d}x}$, so they are linear differential operators. From this *you can check* that L in Eq. (1.31) is a linear differential operator.

Definition 5. *A* **linear ODE** *has the form $Ly = f$ where L is a linear differential operator. If $f = 0$ then it is called a* **homogeneous** *linear ODE; if $f \neq 0$ then it is called an* **inhomogeneous** *linear ODE.*

We have seen examples of linear ODEs in Eq. (1.29) [second order] and Eq. (1.10) [first order]. In the context of physics, inhomogeneous ODEs are sometimes called "forced" equations and homogeneous ODEs are called "unforced" equations. (Take care that *un*forced is homogeneous and forced is *in*homogeneous.)

1.3.2 Principle of superposition

The **principle of superposition** is very important when solving linear ODEs. It has two separate cases

- If y_1 and y_2 are two particular solutions of a **homogeneous** linear ODE then so is $y_1 + y_2$. In fact, $\alpha y_1 + \beta y_2$ is also a solution, where α, β are arbitrary real (or complex) numbers. This can be easily checked, it uses (only) Eq. (1.32).

- Suppose that $y_{\rm p}$ solves an **inhomogeneous** linear ODE ($Ly = f$), and $y_{\rm c}$ is the general solution of the corresponding homogeneous equation ($Ly = 0$). Then the general solution of the inhomogeneous equation is

  $$y = y_{\rm c} + y_{\rm p}$$

  To see that this y is a valid solution, note that

  $$\begin{aligned} Ly &= Ly_{\rm c} + Ly_{\rm p} \\ &= (\text{zero}) + f \end{aligned}$$

  where the first "=" sign uses linearity and the second uses that $y_{\rm c}$ solves the homogeneous equation and $y_{\rm p}$ solves the inhomogeneous equation. This shows that y solves the inhomogeneous equation, as required.[3]

In the second case $y_{\rm p}$ is called the **particular integral** and $y_{\rm c}$ is called the **complementary function**.

The physical idea behind this principle is that different solutions describe different types of behaviour that are somehow independent of each other. Hence we can them up to get new solutions.

1.3.3 Second-order linear ODEs with constant coefficients

For the rest of our analysis of second-order ODEs we restrict to the case where $p(x)$ and $q(x)$ are constants (independent of x). These are called "ODEs with constant coefficients".

Homogeneous equation

We start with the homogeneous equation which is

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + 2p\frac{\mathrm{d}y}{\mathrm{d}x} + qy = 0 \tag{1.33}$$

We assume that p, q are real and we look for solutions where y is real.

To solve this we try a solution of the form $y = \mathrm{e}^{\lambda x}$. We find

$$(\lambda^2 + 2p\lambda + q)\mathrm{e}^{\lambda x} = 0 \tag{1.34}$$

We can't have $\mathrm{e}^{\lambda x} = 0$ so we must solve the quadratic equation $\lambda^2 + 2p\lambda + q = 0$ which gives

$$\lambda = -p \pm \sqrt{p^2 - q}$$

This means that λ may be real or complex. We treat the different cases in turn.

[3](Not examinable) To see that this y is the general solution of $Ly = f$, suppose that $y = y_{\rm p} + y_{\rm x}$ is a particular solution to $Ly = f$. Then $Ly_{\rm x} = L(y - y_{\rm p}) = f - f = 0$. We have shown that $Ly_{\rm x} = 0$ so $y_{\rm x}$ is a particular solution of the homogeneous equation, hence it must be contained in the general solution of that equation, which is $y_{\rm c}$ (by definition). Hence the solution $y_{\rm c} + y_{\rm p}$ contains all solutions to $Ly = f$.

Two real roots, $p^2 > q$. We write

$$\lambda_+ = -p + \sqrt{p^2 - q}, \qquad \lambda_- = -p - \sqrt{p^2 - q}$$

We can solve our ODE as $y(x) = \mathrm{e}^{\lambda_+ x}$ or $y(x) = \mathrm{e}^{\lambda_- x}$. By the principle of superposition (homogeneous case) we can also solve the ODE as

$$y(x) = A\mathrm{e}^{\lambda_+ x} + B\mathrm{e}^{\lambda_- x} \tag{1.35}$$

where A, B are arbitrary real numbers (these are integration constants). We have a second order ODE so we expect our general solution to have two integration constants, from this we can conclude that Eq. (1.35) is the *general solution* for $p^2 > q$.

Two complex roots, $p^2 < q$. We write

$$\Omega = \sqrt{q - p^2}, \qquad \lambda = -p + \mathrm{i}\Omega, \qquad \lambda^* = -p - \mathrm{i}\Omega$$

where the * indicates the complex conjugate. Our trial solutions are $\mathrm{e}^{\lambda x}$ and $\mathrm{e}^{\lambda^* x}$ but these are complex, and we want real solutions. The principle of superposition says that $\frac{1}{2}(\mathrm{e}^{\lambda x} + \mathrm{e}^{\lambda^* x})$ is also a solution, this is in fact the real part of $\mathrm{e}^{\lambda x}$. It gives

$$\frac{1}{2}(\mathrm{e}^{-px}\mathrm{e}^{\mathrm{i}\Omega x} + \mathrm{e}^{-px}\mathrm{e}^{-\mathrm{i}\Omega x}) = \mathrm{e}^{-px}\cos\Omega x \tag{1.36}$$

where we used $\cos\theta = \frac{1}{2}(\mathrm{e}^{i\theta} + \mathrm{e}^{-\mathrm{i}\theta})$. If we take instead the imaginary part of $\mathrm{e}^{\lambda x}$ we get

$$\frac{1}{2\mathrm{i}}(\mathrm{e}^{\lambda x} - \mathrm{e}^{\lambda^* x}) = \mathrm{e}^{-px}\sin\Omega x \tag{1.37}$$

We have two real solutions in Eqs. (1.36) and (1.37). The principle of superposition gives us the general solution

$$y(x) = \mathrm{e}^{-px}[A\cos\Omega x + B\sin\Omega x] \tag{1.38}$$

Note: by writing the two integration constants as $(A, B) = (R\cos\phi, R\sin\phi)$ this can alternatively be written as $y(x) = R\mathrm{e}^{-px}\cos(\Omega x - \phi)$. The general solution can be written in different forms by changing how we represent the integration constants.

Exactly one root, $p^2 = q$. Our trial solution becomes $y(x) = \mathrm{e}^{-px}$. We can't construct a general solution from this alone. However it turns out that there is a second solution, $y(x) = x\mathrm{e}^{-px}$. Check: first note

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{e}^{-px} - px\mathrm{e}^{-px}.$$

Hence

$$\begin{aligned}
\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + 2p\frac{\mathrm{d}y}{\mathrm{d}x} + p^2 y &= \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{-px} - px\mathrm{e}^{-px}) + 2p(\mathrm{e}^{-px} - px\mathrm{e}^{-px}) + p^2 x\mathrm{e}^{-px} \\
&= (-p\mathrm{e}^{-px} - p\mathrm{e}^{-px} + p^2 x\mathrm{e}^{-px}) + (2p\mathrm{e}^{-px} - 2p^2 x\mathrm{e}^{-px}) + p^2 x\mathrm{e}^{-px} \\
&= 0
\end{aligned}$$

as required.

Using our two solutions we finally get the general solution by superposition:

$$y(x) = (A + Bx)\mathrm{e}^{-px} \tag{1.39}$$

We considered three cases (two real roots, two complex roots, one real root). These will be discussed further in Sec. 1.4.

Inhomogeneous equation

We have found the general solution for all cases of the homogeneous equation. Now for the inhomogeneous equation (also known as the forced equation). This is

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 2p\frac{\mathrm{d}y}{\mathrm{d}x} + qy = f(x) \tag{1.40}$$

Remember from the principle of superposition that the general solution of this equation can be obtained by combining the general solution of the homogeneous case (complementary function) with any solution that we can find for the forced case (particular integral).

Our approach to finding particular integrals is to guess a suitable trial solution that contains some parameters, and adjust the parameters until we find something that works. We show this with an example:

Example 8 (Particular integral)**.** *Suppose that the forcing function is $f(x) = F\mathrm{e}^{Kx}$ and write the forced equation as $Ly = f$. As trial solution we take $y = a\mathrm{e}^{kx}$ where a, k are parameters. We find*

$$\begin{aligned} Ly &= k^2 a\mathrm{e}^{kx} + 2kap\mathrm{e}^{kx} + q\mathrm{e}^{kx} \\ &= a(k^2 + 2kp + q)\mathrm{e}^{kx} \end{aligned}$$

To make $Ly = f$ then we must set $k = K$, this gives

$$a(K^2 + 2Kp + q)\mathrm{e}^{Kx} = F\mathrm{e}^{Kx} \tag{1.41}$$

We can solve this by taking $a = \frac{F}{K^2+2Kp+q}$. Hence our particular integral is

$$y_{\mathrm{p}}(x) = \frac{F}{K^2 + 2Kp + q}\mathrm{e}^{Kx} \tag{1.42}$$

Note: if it happens that $K^2 + 2Kp + q = 0$ then we would have to divide by zero so this trial solution does not work. This happens when the K that appears in the forcing is a root of the polynomial Eq. (1.34) that we analysed in the homogeneous case. The problem can still be solved in such cases but it needs a bit more work. [The next step is to try $y(x) = x\mathrm{e}^{Kx}$; if that also fails then try $y(x) = x^2\mathrm{e}^{Kx}$. The examples sheet has an illustration of one such case.]

There are some rules that are helpful when searching for particular integrals (of linear ODEs with constant coefficients):

- If f is a polynomial of degree n (for example $f(x) = Fx^2 + Gx$ has degree $n = 2$) then take as trial solution a general polynomial with the same degree (for example $y(x) = ax^2 + bx + c$).
- If f is an exponential (as in Example 8) then take as trial solution a multiple of that same exponential.
- If f is a linear combination of sines and cosines [including simple cases such as $f = F\sin(Kx)$] then take as trial solution a linear combination of similar sines and cosines (for example $y = a\sin(Kx) + b\cos(Kx)$.)
- If f involves a sum of two cases from the list above (for example $f = x + F\mathrm{e}^{Kx}$) then you can find the particular integrals for the two parts separately and then add them.

  (This last fact follows from the the linearity property in Eq. (1.30): let the forcing be $f_1 + f_2$ the particular integrals corresponding to f_1, f_2 separately be y_1 and y_2. That means $Ly_1 = f_1$ and $Ly_2 = f_2$. But then linearity says that $L(y_1 + y_2) = Ly_1 + Ly_2 = f_1 + f_2$ which means $y_1 + y_2$ is a particular integral for the combined problem.)

Example 9 (General solution)**.** *Take $q = p^2$ and use forcing function $f(x) = F\mathrm{e}^{Kx}$ as in Example 8, with $K + p \neq 0$. Construct the general solution.*

Our equation is

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + 2p\frac{\mathrm{d}y}{\mathrm{d}x} + p^2 y = F\mathrm{e}^{Kx} \tag{1.43}$$

Our complementary function (general solution of homogeneous equation) comes from the case with a single root:

$$y_{\mathrm{c}}(x) = (Ax + B)\mathrm{e}^{-px} \tag{1.44}$$

The particular integral from Example 8 becomes (using $q = p^2$ and factorising the denominator)

$$y_{\mathrm{p}}(x) = \frac{F}{(K+p)^2}\mathrm{e}^{Kx} \tag{1.45}$$

So by principle of superposition (inhomogeneous case) we have general solution

$$y(x) = (Ax + B)\mathrm{e}^{-px} + \frac{F\mathrm{e}^{Kx}}{(K+p)^2} \tag{1.46}$$

where A, B are (arbitrary) constants of integration.

Comparison to 1st order linear ODEs. The principle of superposition applies to all linear ODEs including those of Sec. 1.2.4. For example in the answer to Example 4 [which was Eq. (1.18)] then the particular integral is e^{-2x} and the complementary function is $A\mathrm{e}^{-3x}$.

1.3.4 Boundary conditions and initial conditions

As already noted above, we always have two constants of integration in the general solutions to our second-order ODEs. To get a particular solution we have to fix these constants by using some extra information. There are several ways in which this information might be provided but the two most common cases are:

- We are given the value of the dependent variable and its derivative (for example y and $\mathrm{d}y/\mathrm{d}x$) at some particular value of the independent variable (which in this example would be x). These are usually called **initial conditions** and problems where these conditions are given are usually called **initial value problems**. In fact these more commonly occur in cases where the dependent variable is the time t, see later.

- We are given the value of the dependent variable at two different points. For example, if y is the dependent variable then we might be given that $y(0) = 0$ and $y(1) = 1$. Problems of this type are usually called **boundary value problems**.

Example 10 (Initial value problem)**.** *Solve*

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + 5\frac{\mathrm{d}y}{\mathrm{d}x} + 6y = 3x$$

with initial conditions $y(0) = 1$ and $y'(0) = 0$.

For the particular integral we should try a polynomial $y = ax + b$ which gives

$$5a + 6(ax + b) = 3x$$

This must be true for all x so we compare coefficients of x and 1 to get $6a = 3$ and $5a + 6b = 0$. Solving gives $a = \frac{1}{2}$ and $b = -\frac{5}{12}$. So the particular integral is

$$y_{\mathrm{p}} = \frac{x}{2} - \frac{5}{12}$$

For the complementary function we try $y = \mathrm{e}^{\lambda x}$ in the homogeneous version of the ODE; this gives $\lambda^2 + 5\lambda + 6 = 0$ which means $\lambda = -2$ or -3. Putting together the complementary function and particular integral gives the general solution

$$y(x) = A\mathrm{e}^{-2x} + B\mathrm{e}^{-3x} + \frac{x}{2} - \frac{5}{12}$$

We have $y'(0) = 0$ which means $-2A - 3B + \frac{1}{2} = 0$. Also $y(0) = 1$ which means $A + B - \frac{5}{12} = 1$. Solving the simultaneous equations gives $B = -\frac{7}{3}$ and $A = \frac{15}{4}$. So finally

$$y(x) = \frac{15\mathrm{e}^{-2x}}{4} - \frac{7\mathrm{e}^{-3x}}{3} + \frac{x}{2} - \frac{5}{12}$$

1.4 Example : resonance, transients, and damping

The **damped harmonic oscillator** is a very common physical model that is based on a 2nd order ODE.

The dependent variable is $x = x(t)$, which is the co-ordinate of an oscillating object. The simplest example is a mass on a spring, where x is its position. However, we might also take x as the length of a diatomic molecule, or the amount of deformation of a gong after it is struck. The equilibrium value of the co-ordinate is assumed to be $x = 0$ and there is a restoring force $-kx(t)$ that always pushes it back towards its equilibrium value. There is also a friction force$-g\frac{\mathrm{d}x}{\mathrm{d}t}$ which acts in the opposite direction to the velocity of the object. Here k and g are constants that depend on the physical setup (the "spring constant" and the "friction constant").

In such cases, Newton's second law for this object takes the form

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -kx - g\frac{\mathrm{d}x}{\mathrm{d}t} \tag{1.47}$$

where m is the effective mass of the object, k is the spring constant and g is the strength of the friction. Physical considerations mean that these three numbers should all be positive. For a mass on spring, m is the actual mass; other cases might be more complicated but for this course we ignorer that problem and just focus on solving the equation.

Eq. (1.47) can be rearranged to

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 2\gamma\frac{\mathrm{d}x}{\mathrm{d}t} + \omega_0^2 x = 0 \tag{1.48}$$

where $\gamma = \frac{g}{2m}$ and $\omega_0 = \sqrt{k/m}$. (You may realise that if $\gamma = 0$ then we get back a *simple harmonic oscillator*, whose natural frequency is ω_0.)

Eq. (1.48) is a 2nd order linear homogeneous ODE with constant coefficients. Comparing with the general form of Eq. (1.33) we notice that both coefficients q, p are positive.

1.4.1 Solutions to the homogeneous equation (damping)

Suppose that we are given initial conditions for the position x and velocity $\frac{\mathrm{d}x}{\mathrm{d}t}$ at time $t = 0$, we want to know what happens for $t > 0$.

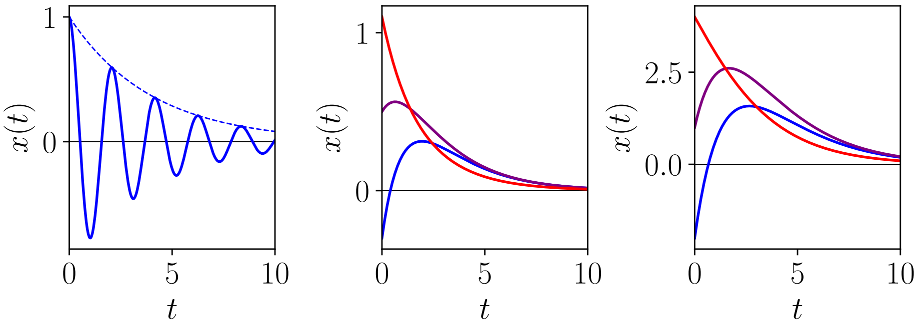

Figure 1.3: Example solutions to the homogeneous equation. (Left) Underdamped case. The dashed line shows $\mathrm{e}^{-\gamma t}$ which sets the rate of decay. (Middle) Overdamped case. (Right) Critical damping.

We can get the solutions by following the method of Sec. 1.3.3. Trying $x(t) = \mathrm{e}^{\lambda t}$ we get

$$\lambda^2 + 2\gamma\lambda + \omega_0^2 = 0$$

and so

$$\lambda = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}$$

As before, there are three cases.

- If $\gamma > \omega_0$ then we have two real roots. The general solution is

  $$x(t) = A\mathrm{e}^{-\left(\gamma-\sqrt{\gamma^2-\omega_0^2}\right)t} + B\mathrm{e}^{-\left(\gamma+\sqrt{\gamma^2-\omega_0^2}\right)t}$$

  Notice that the roots are both negative so these exponentials both decay with time. At long times we have $x(t) \to 0$ so the object returns to its equilibrium position. This case is called **overdamped** (large γ, high friction).

- If $\gamma < \omega_0$ then we have two complex roots. It is convenient to define

  $$\Omega = \sqrt{\omega_0^2 - \gamma^2}.$$

  The general solution is

  $$x(t) = \mathrm{e}^{-\gamma t}\left[A\cos(\Omega t) + B\sin(\Omega t)\right]$$

  This function oscillates as a function of time but it eventually decays to zero, as in the previous case. This case is called **underdamped** (small γ, low friction, long-lived oscillations).

  In the special case $\gamma = 0$ then there is no friction, we get back a simple harmonic oscillator, and the function oscillates forever. This is not usually realistic in physical systems but it is useful as an idealised case.

- $\gamma = \omega_0$ then we have a repeated root and the general solution is

  $$x(t) = \mathrm{e}^{-\gamma t}(A + Bt)$$

  This case is called **critical damping**, it is intermediate between the other two cases.

Examples of all three cases are shown in Fig. 1.3. Note that if there is any amount of friction ($\gamma > 0$) then this "damps" the motion, and the system eventually decays back to its equilibrium position $x = 0$.

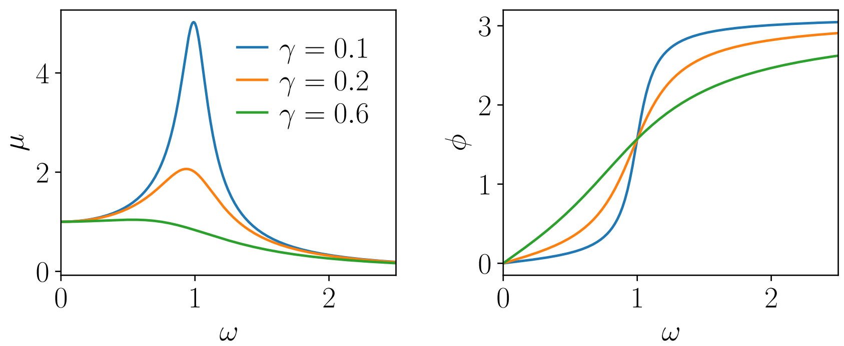

Figure 1.4: Amplitude and phase of the particular integral in the forced case. These are plots of μ and ϕ against ω, with $\omega_0 = 1$ and various values of γ. For small damping γ, we see that μ has a peak near $\omega = \omega_0$.

1.4.2 Sinusoidal forcing (resonance)

Another important physical case occurs when we add an oscillating external force so that our equation becomes inhomgeneous:

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + 2\gamma\frac{\mathrm{d}x}{\mathrm{d}t} + \omega_0^2 x = f_0\cos(\omega t) \tag{1.49}$$

where f_0 is the strength of the forcing and ω is its angular frequency.

We know that this can be solved by a combination of complementary function and particular integral. We have solved the homogeneous equation so we already know the complementary function. For the particular integral we try

$$x_\mathrm{p}(t) = a\cos\omega t + b\sin\omega t \tag{1.50}$$

Putting this into Eq. (1.49) we get

$$-\omega^2[a\cos\omega t + b\sin\omega t] + 2\gamma\omega[-a\sin\omega t + b\cos\omega t] + \omega_0^2[a\cos\omega t + b\sin\omega t] = f_0\cos\omega t \tag{1.51}$$

This must be true for all t so we can compare coefficients of $\cos\omega t$ and $\sin\omega t$. For coefficients of $\sin\omega t$ we get

$$(\omega_0^2 - \omega^2)b - 2\gamma\omega a = 0, \qquad \text{so} \qquad a = \frac{b(\omega_0^2 - \omega^2)}{2\gamma\omega}$$

For coefficients of $\cos\omega t$ we get

$$(\omega_0^2 - \omega^2)a + 2\gamma\omega b = f_0,$$

Substituting for a and doing some rearrangements gives

$$b = \frac{2\gamma\omega f_0}{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}, \qquad a = \frac{(\omega_0^2 - \omega^2)f_0}{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}$$

Putting this into Eq. (1.50) gives the particular integral.

It is useful to remember the general property of sines and cosines, that

$$a\cos\omega t + b\sin\omega t = \sqrt{a^2 + b^2}\cos(\omega t - \phi) \tag{1.52}$$

where $\cos\phi = \frac{a}{\sqrt{a^2+b^2}}$ and $\sin\phi = \frac{b}{\sqrt{a^2+b^2}}$.[4] We obtain

$$x_\mathrm{p}(t) = \mu f_0\cos(\omega t - \phi) \tag{1.53}$$

[4]You may recall that this means $\tan\phi = (b/a)$ but you have to be careful which solution to take for ϕ.

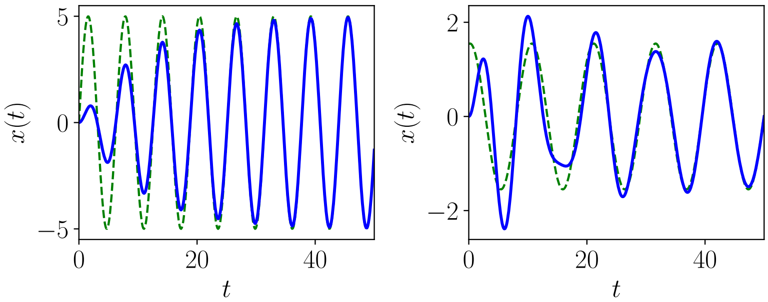

Figure 1.5: Example solutions to the damped harmonic oscillator with $\omega_0 = 1$ with $x(0) = 0$ and $x'(0) = 0$. The blue line is the full solution and the dashed green line is the particular integral. For large times the solution follows the particular integral but there is some transient behaviour at short times. The left panel has $\omega = \omega_0 = 1$ which is the simplest case. The right panel has $\omega = 0.6\,\omega_0$. Values of γ are $0.1\omega_0$ (right) and $0.07\omega_0$ (left).

with

$$\mu = \frac{1}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}}, \qquad \cos\phi = (\omega_0^2 - \omega^2)\mu, \qquad \sin\phi = 2\gamma\omega\mu \tag{1.54}$$

Note that the particular integral oscillates forever, because the system is always being forced (this is different from the complementary function). The size of the oscillation is proportional to f_0, this happens because the ODE is linear. (This can be related back to the principle of superposition.)

The quantity μ determines the amplitude of the oscillations while ϕ tells us whether they are in phase or out of phase with the driving. These quantities are plotted in Fig. 1.4.

The phenomena of **resonance** says that if the damping is small then the response μ has a peak when plotted as a function of the driving frequency ω. This peak occurs when ω is close to the natural frequency ω_0 of the oscillator.

1.4.3 Transients

Remember that the general solution to our ODE is the sum of the complementary function and the particular integral; and that the complementary function includes integration constants which we fix using initial conditions. Some examples are plotted in Fig. 1.5

The key point is that the large-time behaviour of the forced system is an oscillation, given by the particular integral. However, the behaviour for early times depends on the initial conditions, via the complementary function. This early-time behaviour is called a **transient**.